Environment Case Study Gate

Figure 5, Drug Exposure: the Hydrolysate Inhibitor Panels, Held for a Decision

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October 15, 2026

Section status: ✓ ready ■ author-reviewed, keep stable ✗ not done, agents may edit freely

Figure and table status: ● on file, every value in a committed result file ◗ partial, a run unfinished or unscored ❍ idea only, a sketch or nothing run

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1 The decision ✗

Manuscript section R5 covers drug exposure: a compound and its dose as a perturbation the model reads alongside the genotype. The case study is a hydrolysate. Sorghum hydrolysate carries six inhibitors that a fermentation host meets together, and the private Bioscreen runs of 2021 measured all 6 of them on one strain, bAID in aerobic YPD: 180 titration wells, 197 wells over 63 combinations, and 600 wells over 3 isobole grids, 977 wells in all. The models that would predict them were trained on the pooled chemogenomic table: 4 public screens, 6,042,771 cells, 5,463 compounds over 5,863 genes. Nothing here is approved. The sheets are mockups of what the result files support, and the symbol after each panel letter says only whether the values behind it are on file. Color is the source of a predicted value: amber, a measured value, whether a public screen or a Bioscreen well; brick, Bliss independence; lilac, Loewe additivity; wheat, the highest single agent; bronze, the mean of the single-compound profiles; steel blue, a model learned on chemistry features, whose kernel and encoder are written in the label; denim, the cell graph transformer; gray, a gene-mean or permutation control and chance. Hatching is the software growth call and appears only where a panel draws both calls; a plain bar is the served raw-curve call, which is primary. A lighter fill of the same color is a second reading of the same arm, and the panel's legend names it. A label with no bar is an arm that has not run.

What the figure would claim. A model trained on public chemogenomic screens can predict how an industrial host behaves under a mixture of inhibitors it has never seen, and it can do so well enough to rank the combinations a process should avoid. The figure is built to make that claim testable rather than to assert it: the bottom row puts the model beside the two model-free references on the same wells, so a reader sees what the model adds over arithmetic on the single-agent curves, including when the answer is nothing.

The four claims, and where each one stands. First, that growth under a combination departs from independence. On file: over the 63 combinations the Loewe rule reaches a growth AUROC of 0.958 and an accuracy of 0.825, Bliss from the observed singles 0.872 and 0.556, and the highest single agent 0.663; over the 21 combinations that grew, Bliss ranks fitness at Spearman 0.573 with a mean signed error of -0.142 and Loewe at 0.713 with 0.128. Of 15 pairs, 12 are synergistic relative to Bliss, 2 additive and 1 antagonistic. Second, that chemogenomic profile similarity anticipates which pairs deviate. This FAILS as pre-registered: with measured profiles in the matrix the association is 0.204 at a permutation p of 0.462 over 15 pairs, and the all-predicted matrix carries the opposite sign, -0.686 at p 0.006 on raw profiles, which the fingerprint explains rather than the biology. Third, that deletion profiles compose under a mixture by a rule measurable on public data. On file: over 26 Hillenmeyer HET pairs the mean of the singles is the best fixed rule in 24 of them, median R-squared 0.365 against 0.091 for the sum and 0.050 for the highest single, and the free linear fit puts weight 0.529 on one single and 0.593 on the other. Fourth, post-analysis only: composed profiles nominate the genes a screen under each combination would recover, 68 core genes from the best-available matrix and 105 from the all-predicted one. No screen exists to test them.

The bars a model has to clear. Growth: AUROC 0.958, the Loewe rule from the ex21 Hill fits. Fitness over the combinations that grew: Spearman 0.573, Bliss from the observed ex23 singles. Both are model-free, both come from the same wells the model would be scored on, and both are already in a committed result file. On the public side the bar is the nested ridge on fingerprint counts at a median centered Spearman of 0.359 over 96 compound-cold folds, which the next best arm of the ladder trails by a hundredth (0.357).

The angle, after ten rounds. The figure is not a better chemogenomic model. Nested ridge on fingerprint counts holds the public bar (0.359) and no transformer arm of any round reaches it. The angle is that the environment is a perturbation of the same kind as a deletion, and one cell model composes both. Once every record is a cell, a deletion strain in a drug and a wild-type host in a mixture being one record type read through one head, and a mixture is read one inhibitor at a time and composed as Bliss independence, the model trained on the public deletion screens and two anchors orders the 63 private combinations at a growth AUROC of 0.931 without a titration well in training, above the Bliss rule on the measured singles (0.872) and 0.944 with the Hoepfner and Hillenmeyer screens added, at Loewe's level (0.958). The same compositional algebra as the gene arm follows: singles first, with the dose offset giving them a real dose axis (titration Spearman 0.869 at growth AUROC 0.851), then a deviation term, which is the environment's analog of a genetic interaction and the open piece; every dose arm overpredicts the isobole interiors, which is what a missing interaction term looks like. Pooled public data buys transfer, not benchmark points: the screens do not lift the target (0.265 with one source, 0.194 with four, inside one spread) and are not learned at a held-out compound, yet the zero-shot private readout rises with every screen added, three runs per arm and the steps inside the spread, a measured ordering rather than a cleared difference. The private wet-lab data is the environment arm's held-out level, the way the held-out query pairs are the gene arm's: a readout, a culture and a dose regime no training record carries.

Where the full record is. The sheets draw the figure; the measurements behind them are written up elsewhere, and the source table under each figure points to the file. `notes-tex/wet-lab/040-inhibitor-synergy-wetlab` is the case-study document: the 2021 runs, the growth calls, the four claims scored, and the confounds each caption carries. `notes-tex/metabolism/031-betaxanthin-module` and `notes-tex/metabolism/032-amino-acid-module` hold the campaign whose panel conventions this document copies. The chemogenomic side is in three experiment records rather than documents: `experiments/033-env-chemgen-pooled` is the pooled table and its axes, `experiments/035-env-chemgen-vanacloig-cgt` the first compound-cold campaign and the learning curve, and `experiments/038-env-chemgen-vanacloig-cgt-corrected` the ladder and the factorized rounds on the corrected store. The 035 and 038 branches are not landed, so the panel script reads their result files by absolute path and records every path it read in `experiments/040-inhibitor-synergy-wetlab/results/figure_gate_manifest.json`.

What a decision here settles. Whether the figure is the four-by-three of fig. 2 or the three-by-three the experiment note proposes, which of the twelve panels earns main-text space against supplementary space, and whether the bottom row ships with the model beside the rules or with the rules alone. Nothing in this document is a publication-ready panel; the panels are drawn at Nature print width so that the argument, not the artwork, is what gets judged.

2 The mockups ✗

Each caption gives, per panel, what is plotted, the statistic and the status; the table under each figure gives the files behind it. The statistic on the public side is always the per-compound centered Spearman under compound-cold folds, median over compounds, with 26 compounds fitted per fold; on the wet-lab side it is the growth AUROC over the 63 combinations and the fitness Spearman over the 21 that grew under the served call. Color is the source of a predicted value: amber, a measured value, whether a public screen or a Bioscreen well; brick, Bliss independence; lilac, Loewe additivity; wheat, the highest single agent; bronze, the mean of the single-compound profiles; steel blue, a model learned on chemistry features, whose kernel and encoder are written in the label; denim, the cell graph transformer; gray, a gene-mean or permutation control and chance. Hatching is the software growth call and appears only where a panel draws both calls; a plain bar is the served raw-curve call, which is primary. A lighter fill of the same color is a second reading of the same arm, and the panel's legend names it. A label with no bar is an arm that has not run.

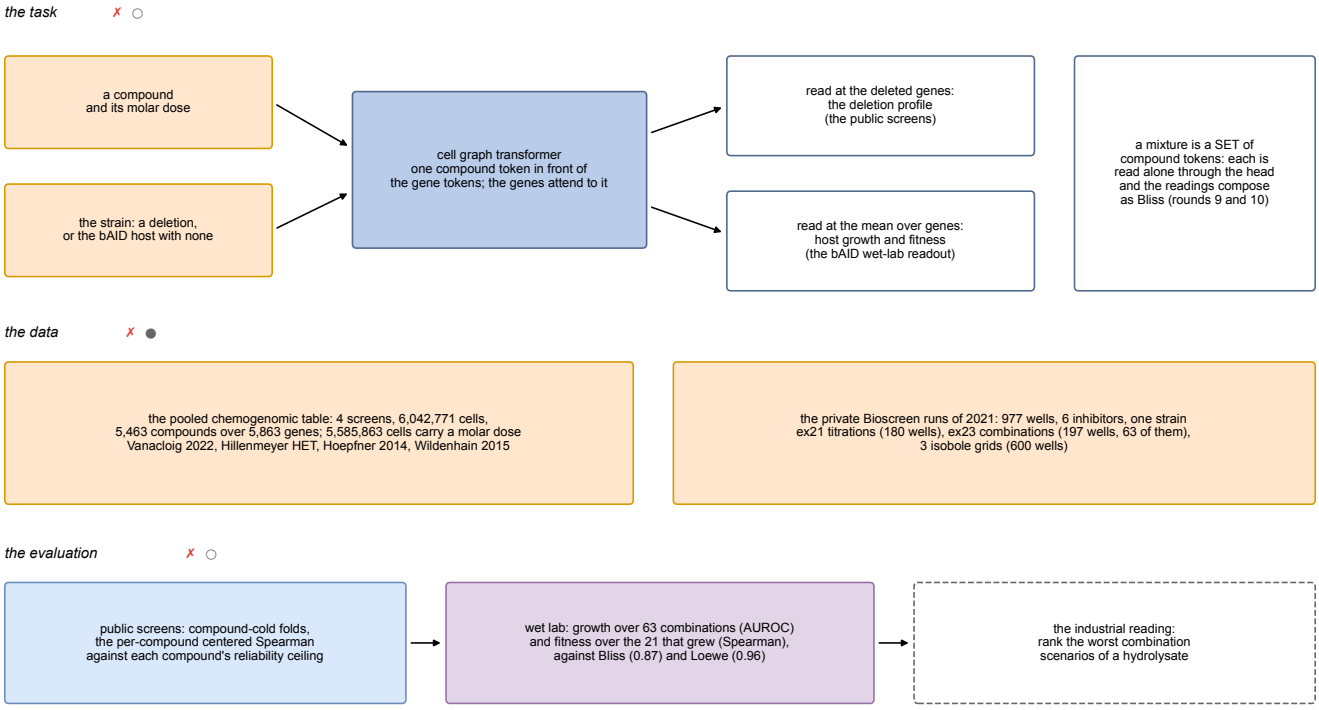


Figure 1: The task, the data and the evaluation. Three blocks, each a layout rather than a drawn result; a dashed box is proposed and not run. **The task** ✗○: a compound and its molar dose enter the cell graph transformer as one token placed in front of the gene tokens, which attend to it; a public-screen strain is read at its deleted genes, giving a deletion profile, and the bAID host, which carries no deletion, is read at the mean over genes, giving growth and fitness. A mixture is a set of compound tokens, one per inhibitor: each is read alone through the shared head and the readings compose as Bliss independence (rounds nine and ten), which is what took the private readout from chance to the rule's level. **The data** ✗●: the pooled chemogenomic table over 4 screens, 6,042,771 cells and 5,463 compounds across 5,863 genes, of which 5,585,863 cells carry a molar dose, beside the private Bioscreen runs, 977 wells over 6 inhibitors on one strain. **The evaluation** ✗○: compound-cold folds and the per-compound centered Spearman on the public side, growth and fitness against Bliss and Loewe on the wet-lab side, and the industrial reading, which is to rank the worst combination scenarios of a hydrolysate.

Sources for fig. 1	
the task	Nothing drawn from a file. The encoder it describes is the environment-encoder head of `038/`; the host head is the proposed adaptation of `experiments/040-inhibitor-synergy-wetlab/scripts/train_mixture.py`. Status: idea.
the data	`033/pool_totals.csv`, `033/store_axes.csv` (the screens, the cells, the compounds, the genes, the molar-dose coverage); `wetlab_wells.csv` and `wetlab_provenance.json` (the wells, the runs, the raw-mirror shas). Status: on file.
the evaluation	`038/ladder/ladder_r2_scores.csv` (the fold size); `mixture_scores.csv` (the two bars). The boxes themselves are a layout, so the block is marked idea even though its counts are on file.

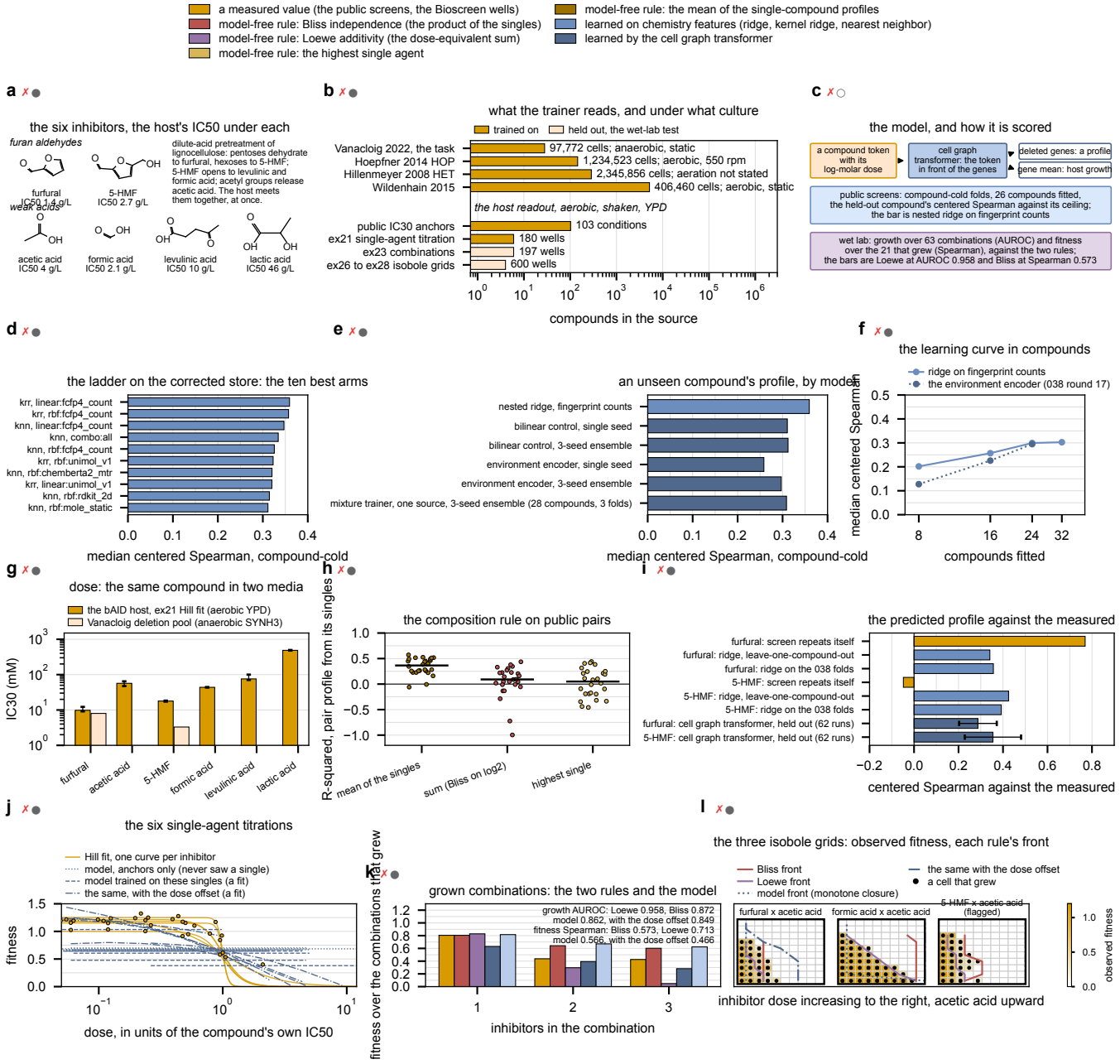


Figure 2: The sheet proposed for the main text, the four-by-three of the experiment note. Columns: the task and the data (left), the public screens (middle), the private wet-lab readout (right). Each panel is described in the paragraph after the figure, with its status.

Panels of fig. 2. **a**, The six inhibitors as skeletal structures from their curated SMILES, in the two families a dilute-acid hydrolysate carries, with the host's half-maximal dose from its titration fit under each. ✗● **b**, What the mixture trainer reads, from the counts a four-source run writes: compounds per source, each bar annotated with its cells, genes and conditions and the culture it was screened in. The task's profiles are 97,772 anaerobic, static cells over 28 compounds; the auxiliary screens add 1,234,523, 2,345,856 and 406,460 aerobic cells, 4,084,611 in all. The host readout is aerobic, shaken YPD, a different oxygen regime from the profiles. The anchors and the titration are trained on; the combinations and the isobole grids never. ✗● **c**, The model in one strip, a compound token with its dose in front of the gene tokens, read at the deleted genes or at the gene mean, and how each side is scored and against what. A layout whose numbers are on file. ✗○ **d**, The ladder, ten best arms of 67: nested ridge on fingerprint counts reads 0.359, the next arm 0.357, the tenth 0.311. ✗● **e**, Which model predicts an unseen compound's profile. The second round is complete: over 96 compound-evaluations, 32 compounds on each of three fold seeds, the three-seed ensemble of the bilinear control reads 0.312 and of the environment encoder 0.297, both below nested ridge at 0.359. Paired on compound and fold, the encoder minus its control is -0.036 with 37 of 96 up. The last bar is the mixture trainer's single-source arm on its own basis (28 compounds with a molar dose, folds 0 to 2, the three-seed ensemble): 0.309. ✗● **f**, The learning curve in fitted compounds. Ridge: 0.202 at eight to 0.303 on the pool, about 0.049 per doubling. The environment encoder (038 round 17, 96 compound-evaluations per point): 0.127, 0.226 and 0.295 at eight, sixteen and twenty-four, below ridge at every count and closing on it. ✗● **g**, Dose, the host's potency against the screen's: furfural 9.82 millimolar against 8, ratio

1.23; 5-HMF 17.9 against 3.30, ratio 5.42; the other four are in no screen. ✗● **h**, How a pair's profile composes from its singles over 26 public pairs: the mean is best in 24, median R-squared 0.365 against 0.091 for the sum and 0.050 for the highest single. ✗● **i**, The predicted profile against the measured one. Furfural: the screen repeats itself at 0.769, ridge reads 0.341; 5-HMF: -0.048, which is noise, against 0.425. The two profiles share 0 significant terms. The cell graph transformer, every round-3 and round-4 run on both machines in which the compound was held out: furfural 0.287 over 62 runs, 5-HMF 0.355 over 62, with the run spread as the whisker. ✗● **j**, The six titrations with their Hill fits, dose in units of each compound's own half-maximal dose; slopes run 1.21 to 20.0. Dotted, the model whose host, the empty genotype read through the shared cell head, trained on the public anchors alone: a flat line at the anchors' assumed fitness for every inhibitor at every dose, Spearman -0.222 against the 54 observed single-agent conditions. Dashed, the model trained on these wells: a fit, Spearman 0.452, and still nearly flat in dose, because without a dose channel the host reads one value per compound. Dash-dot, the same model with the dose offset (round ten): the compound tokens stay dose-free and the dose scales the medium offset, so the curve falls with dose, Spearman 0.890 on these wells. ✗● **k**, The combinations that grew, observed against the two rules, by number of inhibitors. Growth AUROC: Loewe 0.958, Bliss 0.872. Fitness Spearman: Bliss 0.573, Loewe 0.713. The fourth bar is the model (single source, cell readout, trained on the anchors and the titration, mean over 20 runs on 5 folds, seeds and machines), which reads a medium one inhibitor at a time through the shared head and composes the readings as Bliss independence: its predicted fitness falls with the inhibitor count, growth AUROC 0.862, fitness Spearman 0.566. The fifth bar, lighter, is the same model with the dose offset (3 runs): it reads the titration (j) and still composes, growth AUROC 0.849, fitness Spearman 0.466, with the fitness at one inhibitor at the observed level and a shallower fall at two and three. Under mean pooling of the inhibitor tokens the same model read one fitness at every count (AUROC 0.535). ✗● **l**, The 3 isobole grids as observed fitness, white at none, with the cells that grew marked and each rule's front drawn where its predicted fitness crosses the growth threshold: Bliss in red, Loewe in purple. Mean observed minus Bliss over the 81 interior cells: -0.082, -0.446 and -0.017, the last a single run with growth islands and flagged. Dotted, the Bliss-composed model's front after the monotone closure, inside the grid for the first time: the model's mean observed minus predicted over the interior cells is -0.084, +0.016 and -0.102, against the rule's -0.082, -0.446 and -0.017, so on the formic acid grid, where the rule on measured singles overpredicts by 0.45, the model is within 0.02. A caveat the sheet carries: the model's single-agent curves are nearly flat in dose (j), so a product of two low readings matching a low interior is not yet evidence that it reads the interaction; the closure drops 0 cells on the furfural grid. ✗●

Sources for fig. 2

a	`profile_fingerprint_check.csv` (the curated SMILES) and `scripts/mixture_data.py IDENTITY_GAP_SMILES` for formic and lactic acid; `single_agent_fits.csv` for the half-maximal doses. The structures are RDKit 2D coordinates drawn as lines. Status: on file.
b	`results/mixture-delta/r3_vanhophetwil_e150_p0/*_f0_counts.csv`, the per-source counts `scripts/train_mixture.py` writes beside every run; the culture conditions are the loaders' sourced values (`vanacloig2022.py`, `hoepfner2014.py`, `hillenmeyer2008.py`, `wildenhain2015.py`, `private_torchcell/bioscreen.py`; the Bioscreen archive states no shaking, so "shaken" is the operator's statement). Status: on file.
c	`038/ladder/ladder_r2_scores.csv`, `mixture_scores.csv`; the boxes are a layout. Status: idea.
d	`038/ladder/ladder_r2_summary.csv`, centered target, full compound count. Status: on file.
e	`038/ladder/ladder_r2_summary.csv`; `038/factorized/r9_control/*_scores.csv`; `038/factorized/r10_envenc/*_scores.csv`; `results/mixture/r3_van_e150/*_scores.csv` for the mixture-trainer bar. Status: on file.
f	`035/learning_curve/ridge_curve.csv`, centered target; `038/factorized/r17_curve/*_scores.csv` for the encoder. Status: on file.
g	`single_agent_fits.csv` (the host, with bootstrap intervals from `single_agent_bootstrap.csv`); `vanacloig_ic30.csv` (the screen). Status: on file.
h	`het_rule_fit.csv`, `het_summary.json`. Status: on file.
i	`profile_predicted_vs_measured.csv` (`loco_32`, centered); `go_overlap_measured_vs_predicted.csv`; every `results/mixture*/r[34]*/*_vanacloig_compounds.csv` on both machines for the transformer. Status: on file.
j	`wetlab_wells.csv`, `single_agent_fits.csv`, `single_agent_bootstrap.csv`; `results/host_predictions.csv` (`scripts/r3_host_predictions.py`) for the two model curves. Status: on file.
k	`mixture_combinations.csv`, `mixture_scores.csv`, served call; `results/host_predictions.csv` for the model bar. Status: on file.
l	`isobole_cells.csv`, `isobole_summary.csv`; `results/host_predictions.csv` and `mixture_rules.monotone_alive` for the model front. Status: on file.

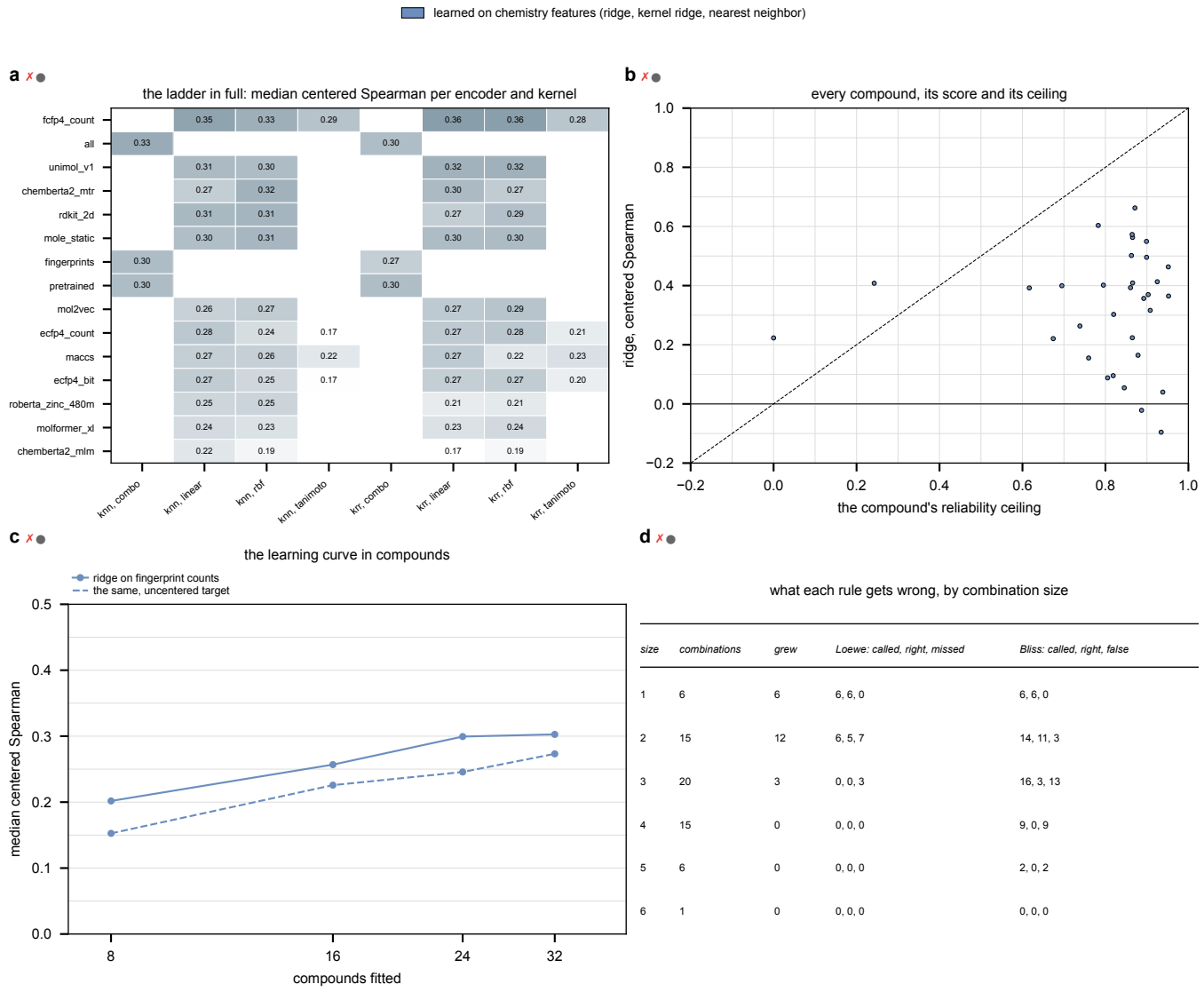
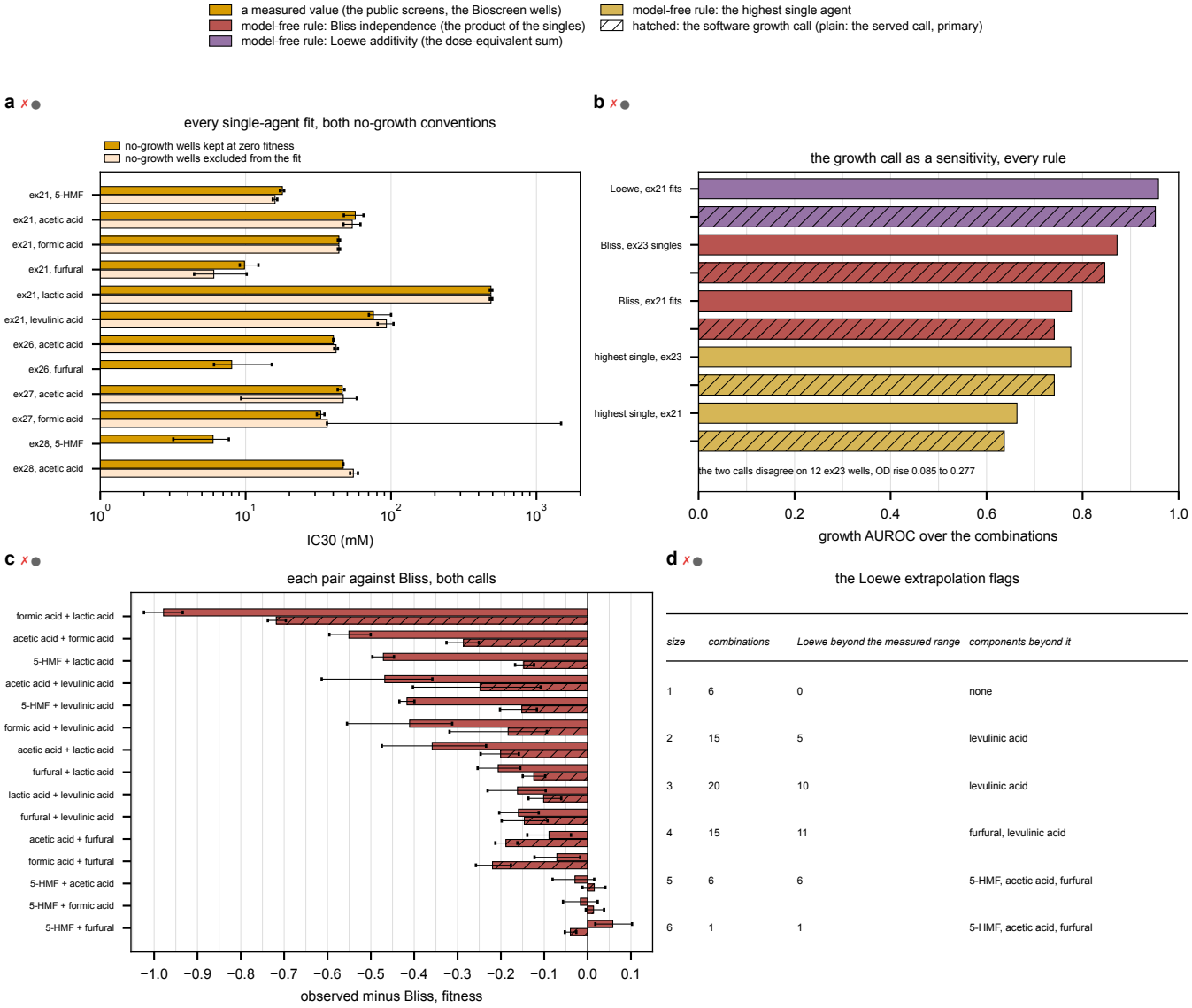


Figure 3: Supplement one: the public benchmark in full. **a**, Every ladder arm, median centered Spearman per encoder and kernel, 62 fitted cells over 15 encoders. Fingerprint counts lead every kernel they are defined for. ✗● **b**, Each of the 32 compounds, its ridge score against its own reliability ceiling: median 0.367, 2 above the ceiling and 2 below zero. ✗● **c**, The learning curve on both targets, the uncentered one dashed. The centered curve is the one the main sheet carries. ✗● **d**, What each rule gets wrong, by combination size: how many combinations the rule called alive, how many of those were right, and what it missed or invented. Loewe misses all 3 growing triples and 7 of 12 growing pairs; Bliss calls 13 of 17 dead triples alive. ✗●

Sources for fig. 3

a	`038/ladder/ladder_r2_summary.csv`. Status: on file.
b	`038/ladder/ladder_r2_scores.csv`, kernel ridge on fingerprint counts, centered, averaged over fold seeds. Status: on file.
c	`035/learning_curve/ridge_curve.csv`. Status: on file.
d	`mixture_scores.csv`, served call, one row per combination size. Status: on file.



size	combinations	Loewe beyond the measured range	components beyond it
1	6	0	none
2	15	5	levulinic acid
3	20	10	levulinic acid
4	15	11	furfural, levulinic acid
5	6	6	5-HMF, acetic acid, furfural
6	1	1	5-HMF, acetic acid, furfural

Figure 4: Supplement two: the single agents and the growth call. **a**, Every single-agent fit, 24 of them over the titration run and the three isobole axes, under both no-growth conventions: the darker bar keeps no-growth wells at zero fitness, the lighter one drops them from the fit. Intervals are the bootstrap. ✗● **b**, Every claim-one score under both growth calls, hatched for the software call. The two calls disagree on 12 wells of the combination run across 4 combinations, with optical-density rises from 0.085 to 0.277. ✗● **c**, Each pair against Bliss with its replicate-bootstrap interval, under both calls. Under the served call 12 of 15 pairs are synergistic, 2 additive and 1 antagonistic. ✗● **d**, Where the Loewe rule had to extrapolate past the measured dose range: 33 of the combinations, and 0 of the isobole fronts fell outside their grid. ✗●

Sources for fig. 4	
a	`single_agent_fits.csv`, `single_agent_bootstrap.csv`. Status: on file.
b	`mixture_scores.csv` (both calls), `wetlab_growth_call_check.csv`. Status: on file.
c	`pair_deviation.csv`. Status: on file.
d	`mixture_combinations.csv` (the extrapolation flags), `isobole_summary.csv`. Status: on file.

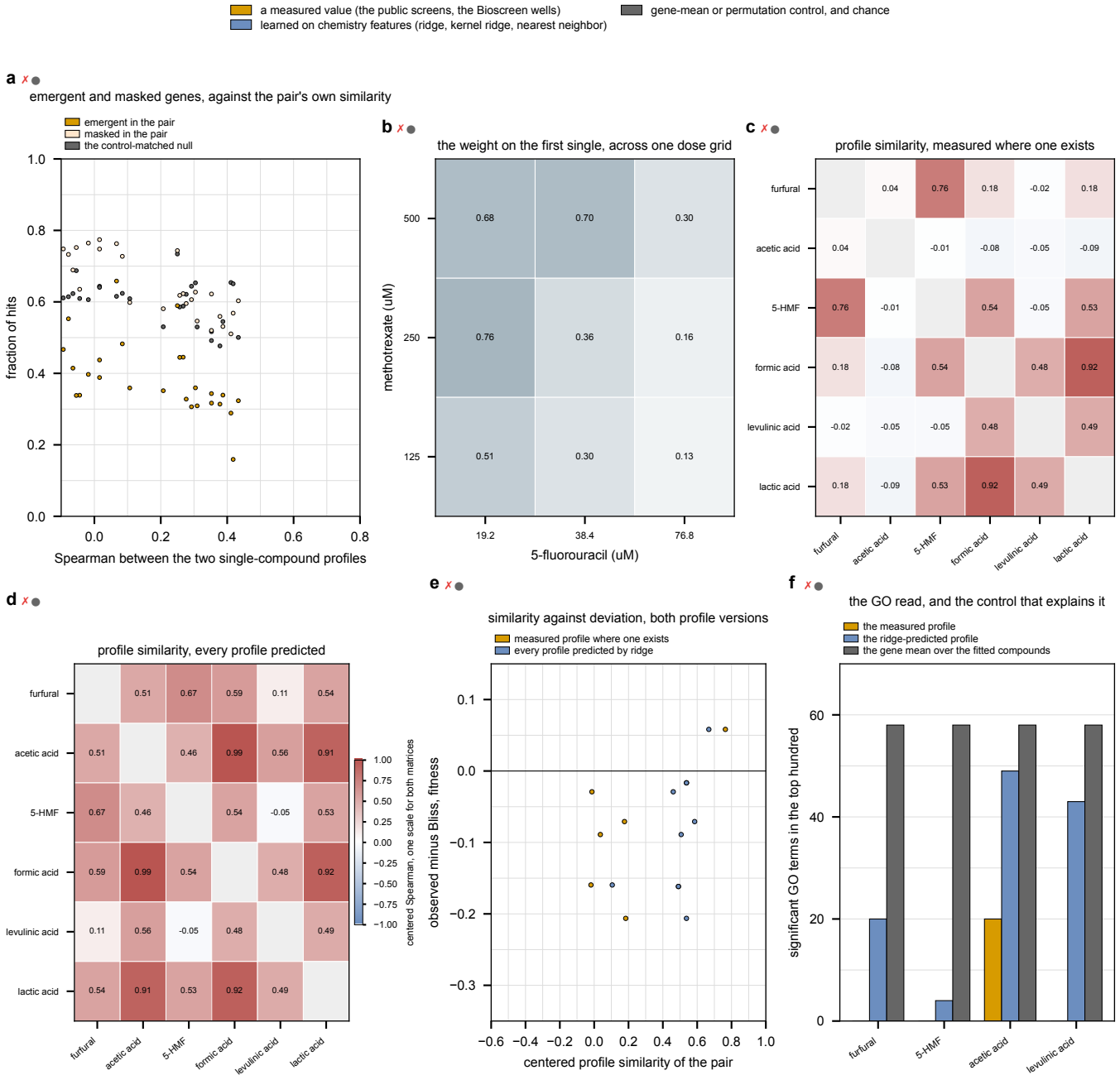


Figure 5: Supplement three: how profiles compose, and what the predicted ones carry. **a**, Emergent and masked gene rates of each public pair against how alike its two singles are, with the control-matched null beside them. The median emergent fraction is 0.355 against a null of 0.621, and the masked fraction 0.649; the only two near-replicate singles in the screen disagree on 0.536 to 0.603 of their own hits, which is the noise floor these rates sit inside. ✗● **b**, The one dose grid the public screen holds, methotrexate against fluorouracil: the weight the free linear fit puts on the first single moves from 0.128 to 0.757 across 9 dose pairs, while the mean wins at 9 of them. ✗● **c**, **d**, Profile similarity among the six inhibitors, measured where a measurement exists and then with every profile predicted. Measured, the strongest pair reads 0.924; predicted, 0.986, because the small acids have near-identical predicted profiles. One scale carries both panels. ✗● **e**, Similarity against the pair's departure from Bliss, both versions: 0.204 at a permutation p of 0.462 with measurements in, and -0.225 at 0.425 with everything predicted. ✗● **f**, The GO read: significant terms in the top hundred sensitive genes of the measured profile, of the ridge-predicted profile, and of the gene mean over the fitted compounds. The control carries 58 terms and the predicted profiles share 48 of them, which is what the predicted enrichment is. ✗●

Sources for fig. 5	
a	`het_emergent_masked.csv`, `het_near_replicates.csv`, `het_summary.json`. Status: on file.
b	`het_rule_fit.csv`, the methotrexate and fluorouracil family. Status: on file.
c, d	`profile_similarity.csv`, centered target, both versions. Status: on file.
e	`similarity_vs_deviation_pairs.csv`, `similarity_vs_deviation.csv`. Status: on file.
f	`go_overlap_measured_vs_predicted.csv`, `nominations_summary.csv`. Status: on file.

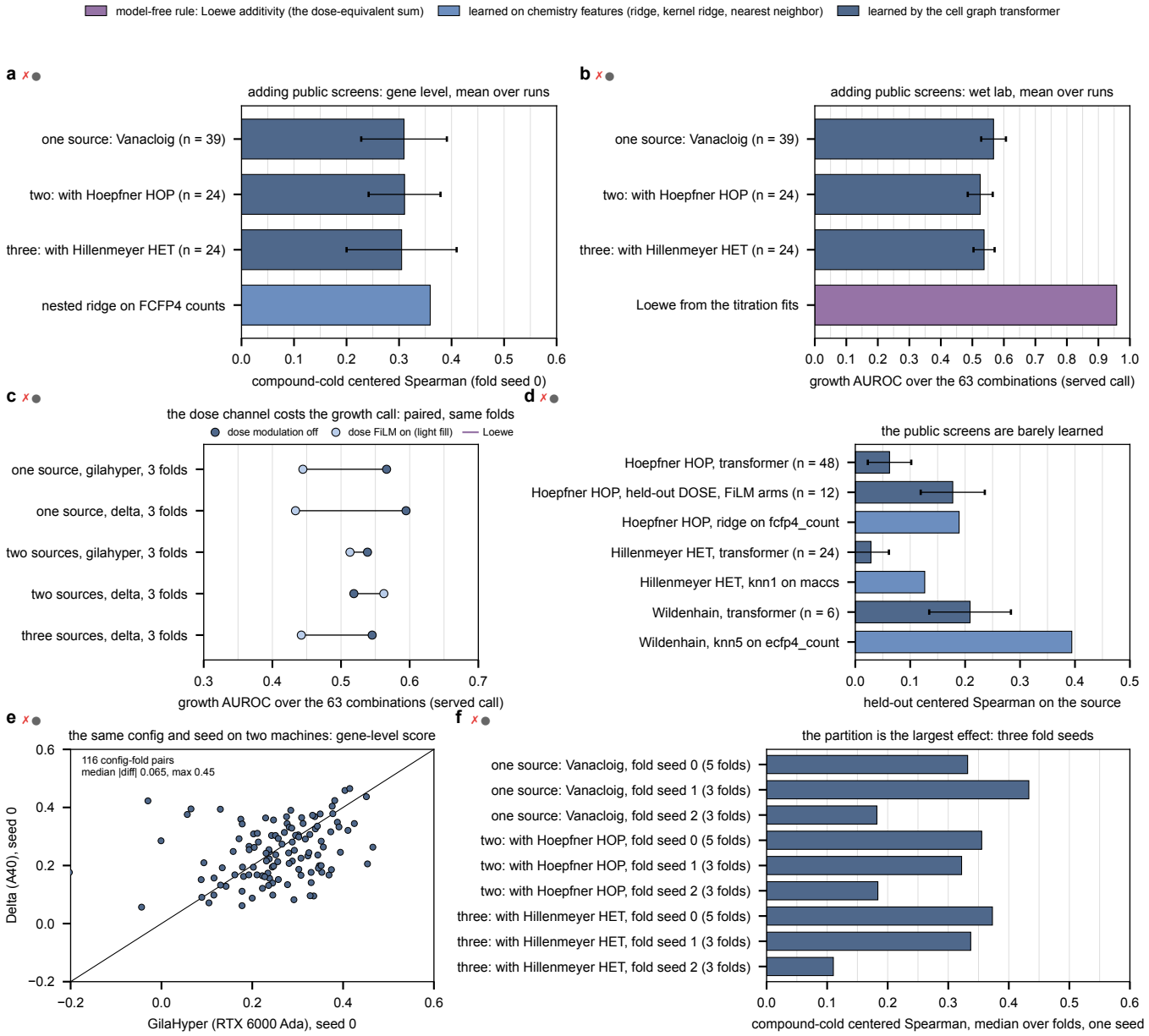


Figure 6: Supplement four: what the mixture rounds say, read over both machines. Rounds three and four of the mixture trainer, every run on fold seed 0 unless a panel says otherwise; a run is one seed on one fold on one machine, and an arm run on both machines contributes both copies. **a**, The one-, two- and three-source arms on the public side, mean over runs with its standard deviation, 39, 24 and 24 runs: 0.310, 0.311 and 0.305, against nested ridge at 0.359. The run spread is 0.082 to 0.105. ✗● **b**, The same runs on the wet lab, growth AUROC over the 63 combinations under the served call: 0.567, 0.525 and 0.537, against Loewe at 0.958. These are the gene readout's arms; the same ladder under the Bliss cell readout, at three seeds, is fig. 10. ✗● **c**, The dose channel, paired: the same arm with the log-molar dose modulation on and off, same machine, same folds, one seed; 5 pairs, FiLM lower on 4 of them, and the furfural-grid residual more negative on 5 of them. ✗● **d**, What the transformer predicts on the public screens it learns from: the held-out compound score per source, mean over the plain arms that carry it, beside the chemistry-feature learner's compound-cold score on the same source (Hoepfner 0.062 against 0.189, Hillenmeyer 0.029 against 0.126, Wildenhain 0.209 against 0.394); and the held-out DOSE of a seen Hoepfner compound under FiLM, 0.177. ✗● **e**, The same config at the same seed on the two machines, every round, 116 config-fold pairs: median absolute difference 0.065, largest 0.452, Spearman between the copies 0.332. That is the run-to-run floor at a fixed seed. ✗● **f**, The partition: each source-set arm on three fold seeds, one seed, median over the folds on file. The single-source arm reads 0.332, 0.433 and 0.182. ✗●

Sources for fig. 6	
a, b	every `results/mixture/r3*/*_scores.csv` and `r4*/*_scores.csv` on GilaHyper and `results/mixture-delta/r3*/*_scores.csv` pulled from Delta, seed members of the arms `mix_van_e150`, `mix_vanhop_e150` and `mix_vanhophet_e150`; the ridge bar from the 038 ladder, the Loewe bar from `mixture_scores.csv`. Status: on file.
c	the same files, the `_film` arm against its dose-off twin on the folds both have, seed 0. Status: on file.
d	the same files, rows whose task is the auxiliary source (`heldout_compound`, `heldout_dose`); the chemistry-feature bars from `source_basis.csv`. Status: on file.
e	the same files, every config and fold scored on both machines at seed 0. Status: on file.
f	the same files, the `_fs1` and `_fs2` arms. Status: on file.

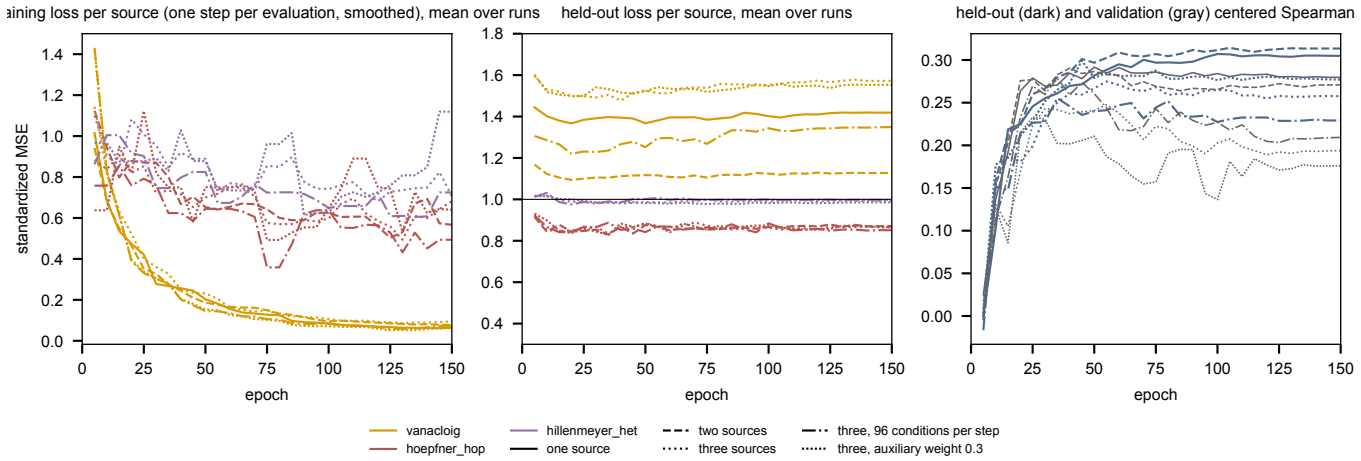


Figure 7: Supplement five: is the trainer converged, per source. Mean curves over every round-3 and round-4 run on both machines of the one-, two- and three-source arms and two three-source variants; the learning rate is zero at epoch 150 on every arm. **a**, The training loss per source, one step per evaluation, smoothed. The target's falls 0.545 to 0.062; Hoepfner's 0.865 to 0.573; Hillenmeyer's RISES, 0.869 to 0.824. ✗● **b**, The held-out loss per source. The target's removes 0.144 of its first value, bottoms near epoch 54 and rises 0.125 by the end; Hoepfner's removes 0.102; Hillenmeyer's 0.043; Wildenhain's 0.449, bottoming near epoch 100 and still falling at 150. ✗● **c**, The held-out and validation centered Spearman by epoch: the held-out score plateaus near epoch 60 to 75 while the held-out loss has already turned, so the ranking holds as the scale drifts. ✗●

Sources for fig. 7	
a to c	every `results/mixture/r3*/*_history.csv` and `r4*` on GilaHyper and `results/mixture-delta/r3*/*_history.csv` from Delta, read by `scripts/r3_convergence.py` into `results/r3_convergence_runs.csv` and `r3_convergence_arms.csv`. Status: on file.

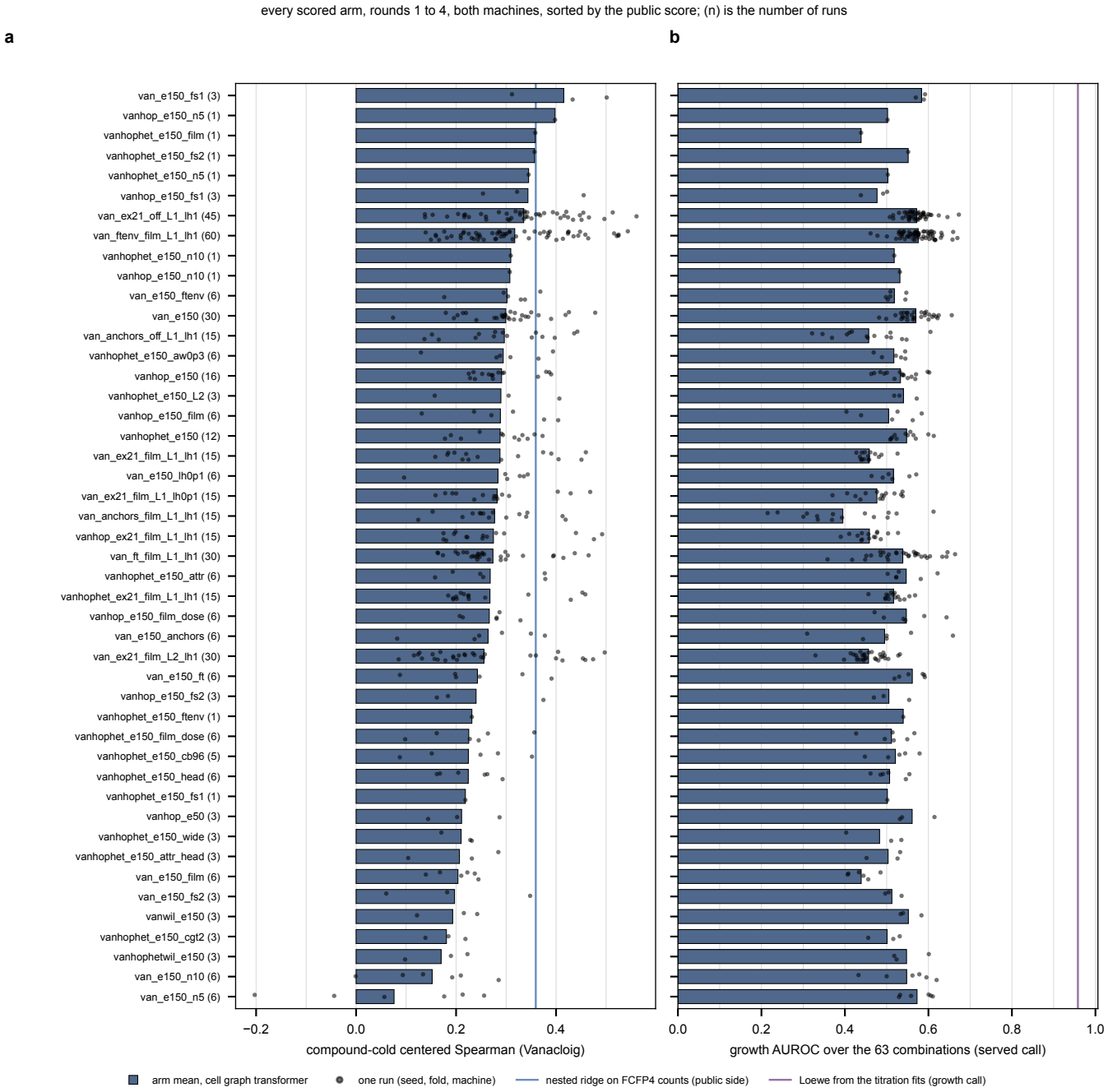


Figure 8: Supplement six: every scored arm of rounds one to four, both machines. One row per arm, every run a point (one seed on one fold on one machine) and the arm's mean a bar, sorted by the public score; rounds one and two (pool-and-last, 50 epochs) sit on the same axis as rounds three and four (the held-out protocol, 150 epochs). **a**, The compound-cold centered Spearman on Vanacloig against nested ridge at 0.359. **b**, The growth call over the 63 combinations against Loewe at 0.958. The spread of the points within an arm against the range of the arm means is the noise floor: no arm clears ridge and none approaches Loewe. ✗●

Sources for fig. 8	
a, b	every `*_scores.csv` under `results/mixture/` and `results/mixture-delta/`, drawn by `scripts/r3_plots.py`; the reference values from `results/figure_gate_numbers.json`. Status: on file.

2.1 The encoders, and how they were compared

Every encoder in this document was scored on ONE basis, so the sheet's bars are comparable: a compound-cold split of the Vanacloig profiles (the compound is held out with every one of its doses; 32 compounds on each of three fold seeds, 96 compound-evaluations), the centered Spearman across strains between the predicted and the measured profile of the held-out compound, and the median over the evaluations as the score. Three kinds of encoder sit on that basis.

- **Fixed chemistry encoders with a shallow learner** (the ladder, 038): 15 ways of embedding a compound, four fingerprint families (FCFP4 and ECFP4 as counts or bits, MACCS keys), RDKit's 2D descriptors, seven pretrained molecular models (mol2vec, MolE, Uni-Mol, ChemBERTa-2 in both pretraining modes, MoLFormer-XL, a ZINC RoBERTa) and three concatenations, each run with kernel ridge and nearest neighbors under linear, RBF and Tanimoto kernels, 62 learner-by-encoder arms, every regularization chosen by a nested fold. The best is nested ridge on FCFP4 counts at 0.359; the best pretrained model is Uni-Mol v1 (3D) at 0.323; the lowest single encoder is ChemBERTa-2 MLM at 0.219.
- **Learned environment encoders** (038 rounds 9 and 10): the bilinear control (a compound embedding read against a learned gene embedding, no attention) and the environment encoder (the compound token with its dose in front of the gene tokens of the cell graph transformer), trained on the same folds, three seeds each; the encoder is read against its control paired on compound and fold, median difference -0.036, up on 37 of 96.
- **The mixture trainer** (040), the same transformer with the public screens and the host records in one objective, which scores on its own basis (the 28 compounds with a molar dose, folds 0 to 2): 0.309 for the gene readout and, under the cell readout, inside its fold noise (section 4).

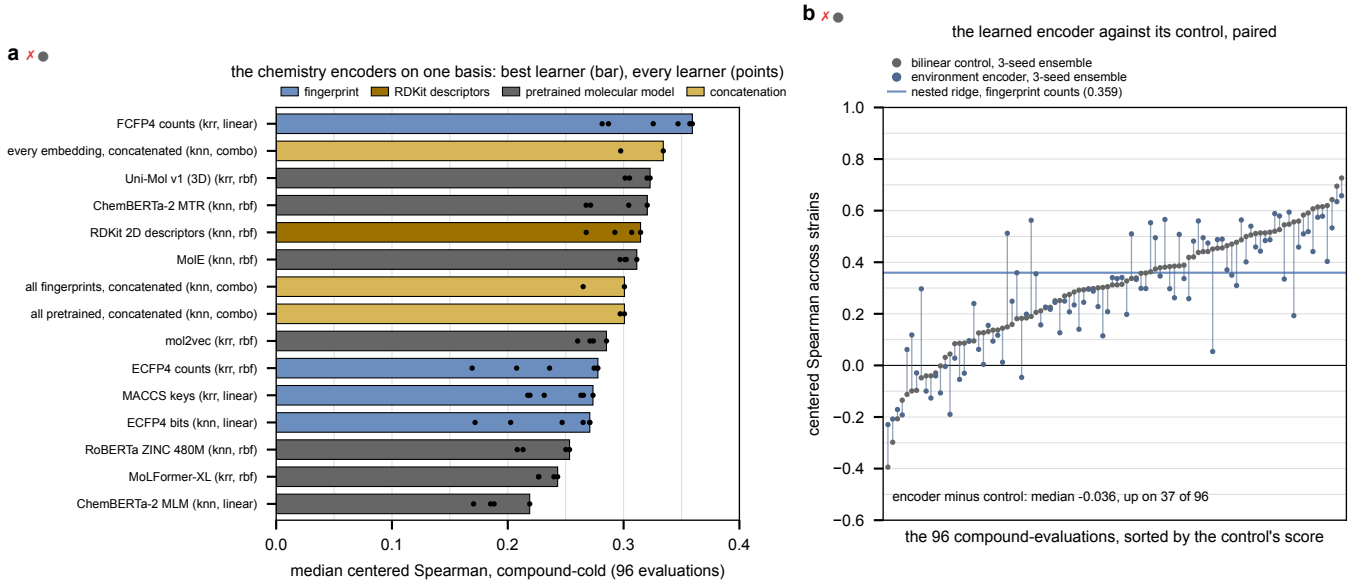


Figure 9: Supplement seven: the encoders compared. a, Every chemistry encoder of the ladder on the one basis, its best learner as the bar (the learner named in the label) and every learner it was run with as a point; color is the encoder family: fingerprints, descriptors, pretrained models, concatenations. Nested ridge on fingerprint counts leads; no pretrained model reaches it. **b**, The learned environment encoder against its bilinear control, paired on compound and fold over the 96 evaluations, sorted by the control's score; the line is nested ridge. The encoder is not above its control (median difference -0.036) and neither is above ridge. ✗●

Sources for fig. 9

a	`038/ladder/ladder_r2_summary.csv`, the centered target at the full compound count, every learner-by-encoder row. Status: on file.
b	`038/factorized/r9_control/*_scores.csv` and `038/factorized/r10_envenc/*_scores.csv`, the three-seed ensemble per fold, merged on compound, fold and fold seed. Status: on file.

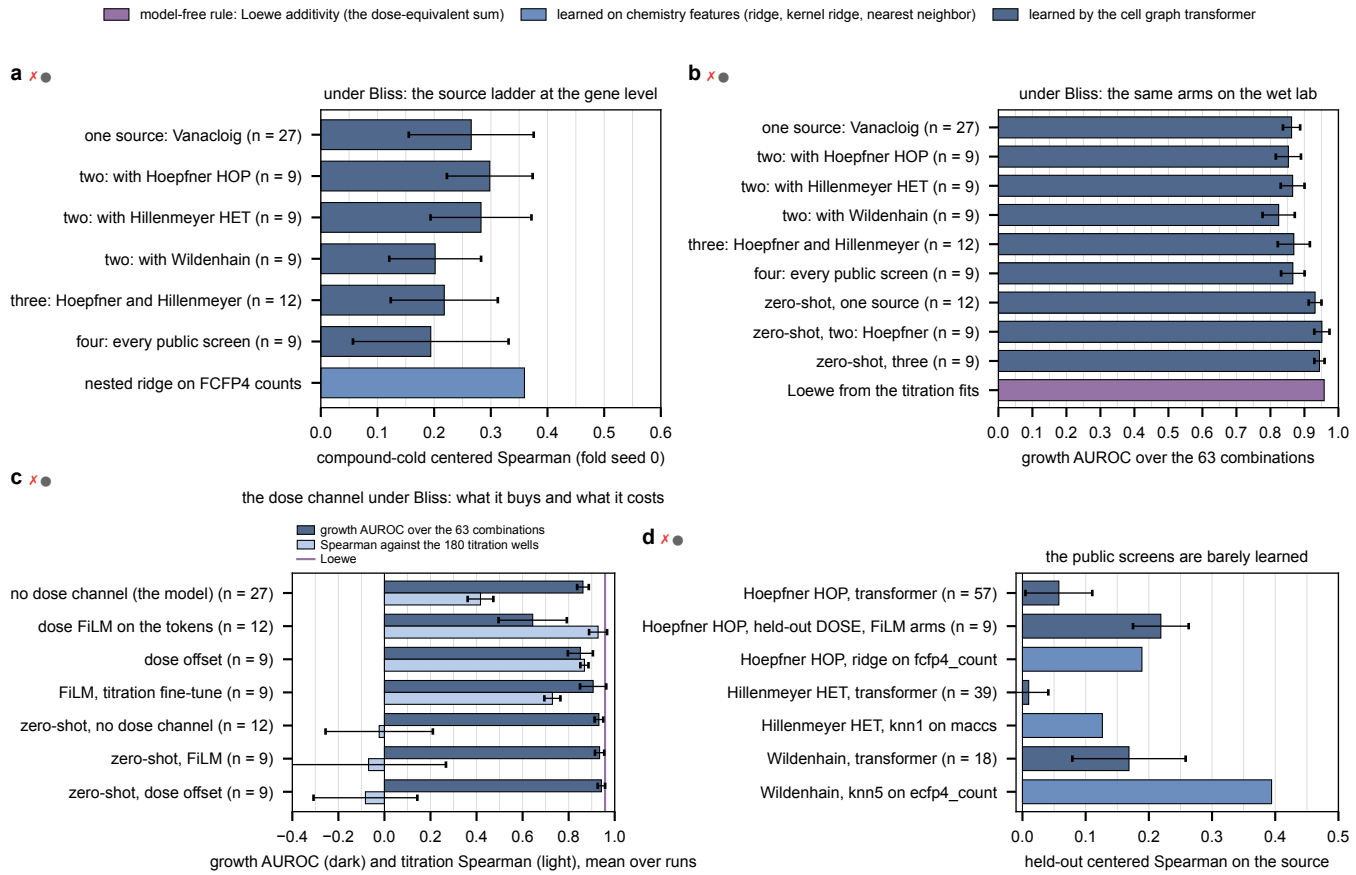


Figure 10: Supplement eight: round ten, the Bliss cell readout at three seeds. Every arm is three seeds on the three screening folds of fold seed 0, read on both machines where it ran on both; the bar is the mean over runs, the whisker the standard deviation. **a**, The source ladder on the public side: one source 0.265, with Hoepfner 0.298, with Hillenmeyer 0.283, with Wildenhain 0.202, three 0.218, four 0.194, run spreads 0.110 to 0.137, against nested ridge at 0.359. No public screen lifts the target, and the order is more sources, lower, inside one spread. ✗● **b**, The same arms on the wet lab, growth AUROC over the 63 combinations: the fitted ladder reads 0.862 to 0.866 and does not move with sources; the zero-shot ladder (anchors only, never a titration well) reads 0.931, 0.951 and 0.944 on one, two and three sources, every value above the Bliss rule on the measured singles (0.872) and the last at Loewe's (0.958). Three runs per arm; the steps are inside one spread, so this is a measured ordering rather than a cleared difference. ✗● **c**, The dose channel: each arm's growth AUROC (dark) beside the Spearman of its host prediction against the 180 titration wells (light), a fit for the arms trained on the titration and a prediction for the zero-shot arms. FiLM on the tokens reads the titration at 0.928 and composes at 0.644; the dose offset reads it at 0.869 and composes at 0.851, against 0.862 with no channel; the titration fine-tune with FiLM reads 0.729 and composes at 0.906. Zero-shot, no channel reads the titration, and the dose offset composes at 0.942. ✗● **d**, What the Bliss arms predict on the public screens they learn from, the held-out compound score per source over every round-ten arm that carries it: Hoepfner 0.058 against 0.189, Hillenmeyer 0.010 against 0.126, Wildenhain 0.169 against 0.394; the held-out Hoepfner dose under FiLM, 0.219. The screens are not learned at a held-out compound, so a screen could not have helped the target at one. ✗●

Sources for fig. 10

a, b	`results/mixture-delta/r10*/*_scores.csv` pulled from Delta and `results/mixture/r9*/*_scores.csv` on GilaHyper (the seed-0 copies of the one- and three-source Bliss arms), seed members on fold seed 0; the ridge bar from the 038 ladder, the Loewe bar from `mixture_scores.csv`. Status: on file.
c	the same files, the `ex23` growth AUROC and the `ex21` all-wells fitness Spearman of the seven dose arms. Status: on file.
d	the same files, rows whose task is the auxiliary source (`heldout_compound`; `heldout_dose` of `mix_vanhop_e150_cell_bliss_film_dose`); the chemistry-feature bars from `source_basis.csv`. Status: on file.

3 What each panel still needs ✗

Table 1: What each panel still needs before it is real. The status column repeats the panel's mark in words: on file (filled circle), every drawn value is in a committed result file; partial (half circle), a run is unfinished or a planned arm has no file; idea only (open circle), a layout or nothing run. No panel is approved; approval is the section status and is the author's.

panel		what remains
concept; main, a	○	Redraw as one draw.io composition at print size. The counts in it are on file; the boxes and arrows are a layout, and the mixture path they describe (a set of compound tokens, each modulated by its own log-molar dose) is the proposed model and has not run.
main, c	○	Same: a layout whose numbers are on file. It merges into the concept sheet if the figure is cut from four rows to three.
main, e; SI-1, a	◗	The second factorized round is producing. Two folds of the bilinear control and two of the environment encoder were on disk when the sheet was drawn; the round's remaining folds, the nine-seed ensembles and the stack with nested ridge are not written yet. Round eleven on Delta adds the width and depth axis.
main, f	◗	The encoder arm of the learning curve has to be rerun on the corrected store; only the ridge arm has a file. The planned round also adds the other pooled sources to the gene-level task, which is the second line this panel would carry.
main, i	◗	The transformer's compound-cold profiles for furfural and 5-HMF come out of the same running round. Until then the panel shows ridge against the measured profile and the screen's own repeat, and the transformer row is a label with no bar.
main, j	◗	The host head's predicted single-agent curves now exist for the anchors-only arm (zero-shot) and the arms that train on the titration run (section 5); the panel is not yet drawn from them.
main, k, l	◗	Round-one model predictions over the combinations and the isobole grids are on file for 5 arms and below the rules on every readout (table 2); the boundary scorer of section 5 has not yet been run on the model surfaces. The panels ship with the rules and the model side by side and the text says so, unless a queued arm clears a bar.
boundary, a–f; rule choice, a, b	●	Nothing. Both figures of section 5 are drawn by committed scripts from committed result files. Open: whether the boundary overlap replaces growth AUROC as the primary wet-lab statistic.
main, b, d, g, h	●	Nothing. Every drawn value is in a committed result file. These four are what the figure can claim today.
SI-1, b, c, d	●	Nothing.
SI-2, all	●	Nothing.
SI-3, all	●	Nothing, with one caveat the caption carries: the similarity matrices mix a measured profile for two inhibitors with predicted profiles for four, and the all-predicted version exists so that mixture is visible rather than hidden.

The running and planned rounds, and what each one fills. The second factorized round of the corrected-store campaign is on the two GilaHyper cards named for it, one job for the bilinear control and one for the environment encoder; it fills main **e**, main **i** and the first column of SI-1 **a**. Round eleven of the same campaign goes to Delta under the account the plan names, never a preemptible partition, and widens the ladder rather than deepening it. The first mixture round of this experiment is the one that fills the bottom row: about ten arms by three seeds, crossing the pooled sources against the host-training recipe (the public dose anchors alone, the anchors plus the titration run, and a fine-tune on the titration run) against the dose treatment. It starts once the mixture trainer passes its processor smoke test, and its launchers are the pair of scripts the experiment folder already carries.

The cut from four rows to three. The experiment note proposes merging rows two and three. The merge that costs least is to move main **c** into the concept sheet, where the same boxes already appear, and to put main **f** and main **g** side by side as one panel on two axes, since both are about what more data buys: more compounds on one axis, a second medium on the other. That leaves nine panels: the task, the data, the ladder, the transformer against ridge, the merged data-and-dose panel, the composition rule, the predicted profile, the single agents, and the combinations with the isoboles as its inset. The alternative, dropping the single-agent panel, is worse: it is what every rule in the bottom row is computed from, and a reader who cannot see the Hill slopes cannot judge why Bliss and Loewe disagree as far as they do.

4 State on the day the sheets were drawn ✗

What had finished. The processor round of this experiment: the four pre-registered claims scored, which is every panel marked on file above. The ladder of the corrected-store campaign, giving main **d** and SI-1 **a**. The learning curve of the earlier campaign, giving main **f** in its ridge arm. The pooled table and its axes, giving main **b**.

What finished while the sheets were drawn. The second factorized round of the corrected-store campaign, submitted in the small hours of the same day on two GilaHyper cards: one job for the bilinear control, one for the environment encoder, each fifteen configurations of five compound-cold folds on three fold seeds, three seeds per fold. Both completed, giving main **e** in full: the control's three-seed ensemble at 0.312 and the encoder's at 0.297 over 96 compound-evaluations, against nested ridge at 0.359, with the encoder -0.036 against its own control when the two are paired on compound and fold. Neither head reaches the fingerprint ridge on this store, which reverses the earlier campaign's reading and is the result the two Delta rounds below are meant to explain or confirm. The sheet re-reads the directory every time it is drawn, so a later round replaces these values rather than adding to them.

What had not started. The first mixture round, which is every panel of the bottom row in its model arm. The trainer and its configuration groups are owned by the implementation thread and are not part of this document.

What to expect, labeled. Hypothesis (untested): the environment encoder ties nested ridge on the corrected store, as it did on the earlier store, so main **e** closes with the two within a hundredth of each other and the figure's public-benchmark row says that architecture is not where the gain is. Hypothesis (untested): the encoder's compound-cold profiles for the two inhibitors in store read no better than ridge, so main **i** keeps its shape and the predicted profiles stay generic. Hypothesis (untested): a dose-aware host head fitted on the titration run beats the highest single agent on growth, which reads 0.663, and does not beat Loewe, which reads 0.958, because Loewe already uses the fitted single-agent curves and the model has no information about this host that those curves do not carry. If that last one holds the honest figure is the model-free one, and the bottom row says so in the caption rather than in a footnote.

What would change the figure's claim. A model arm that beats 0.958 on growth, or 0.573 on fitness over the combinations that grew, moves the bottom row from a validation of arithmetic to a result about transfer. A model arm that does neither still earns the row, because the comparison is the point: the rules are computed from six titrations on the same strain, and a model trained on 5,463 public compounds that cannot beat them is the measurement the section needs to report.

The same day, evening. The first mixture round ran: two arms on the GilaHyper cards named for this work, two on Delta, two more on GilaHyper by the evening, and the rest on Delta overnight; table 2 holds the arms with files here. The third hypothesis above held in its pessimistic half: no arm beats Loewe on growth or Bliss on fitness, and the dose-aware arms do not beat the highest single agent either. The readout was then reframed around the alive and dead boundary (section 5), which is model-free and splits by pair, and the leave-one-pair-out test of rule-choice predictors was added as the bar a learned model must clear. Round two began with the replication of the dose-off joint arm on two further fold seeds.

Three days on, 2026.10.15. Ten rounds are on file, every arm on both machines. The readout is the cell, the composition is Bliss, the model sits at the rules' level zero-shot on the private combinations (0.931 with one source, 0.944 with three) and the dose offset gives it a dose axis at no public cost; the public bar is still ridge's. The bottom row ships with the model beside the rules, which was the third open decision. What is not on file: a deviation term beyond Bliss, a gene-cold split on the public screens, and any arm above ridge on the public side.

5 The boundary readout, and what the first model arms say ✗

For several inhibitors the useful prediction is the shape in dose space inside which the host still grows, the line between alive and dead at each concentration. The fixed-dose combination plate gives one point per combination, so the boundary lives in the three isobole grids and in the single-agent curves that fix where it meets the axes. Bliss multiplies surviving fractions, two independent hits; Loewe adds doses as fractions of each compound's own dose for the same effect, one shared capacity. With the steep single-agent curves of this host the two fronts sit far apart, so which one a pair follows is a large, visible difference, and it is scored here directly: a rule's alive region is the interior cells where its prediction clears the grid's growth threshold, compared with the observed alive region by the overlap of the two sets (the Jaccard index), the cells wrong in each direction, and the growth edge, the highest dose called alive in each column, in grid steps.

Post-processing the observed region. A dose-response boundary is monotone: more inhibitor never rescues growth. A grown cell sitting above a dead one is therefore a call error, and the observed region is closed downward before it is scored (a cell stays alive only if every cell with lower doses on both axes is alive). The furfural and formic acid grids are already monotone. The 5-HMF grid, flagged since the processor round for its growth islands, loses 4 interior cells to the closure, going from 17 alive cells to 13; the figure marks each dropped cell with a cross. Any predicted surface gets the same closure before it is scored as a boundary.

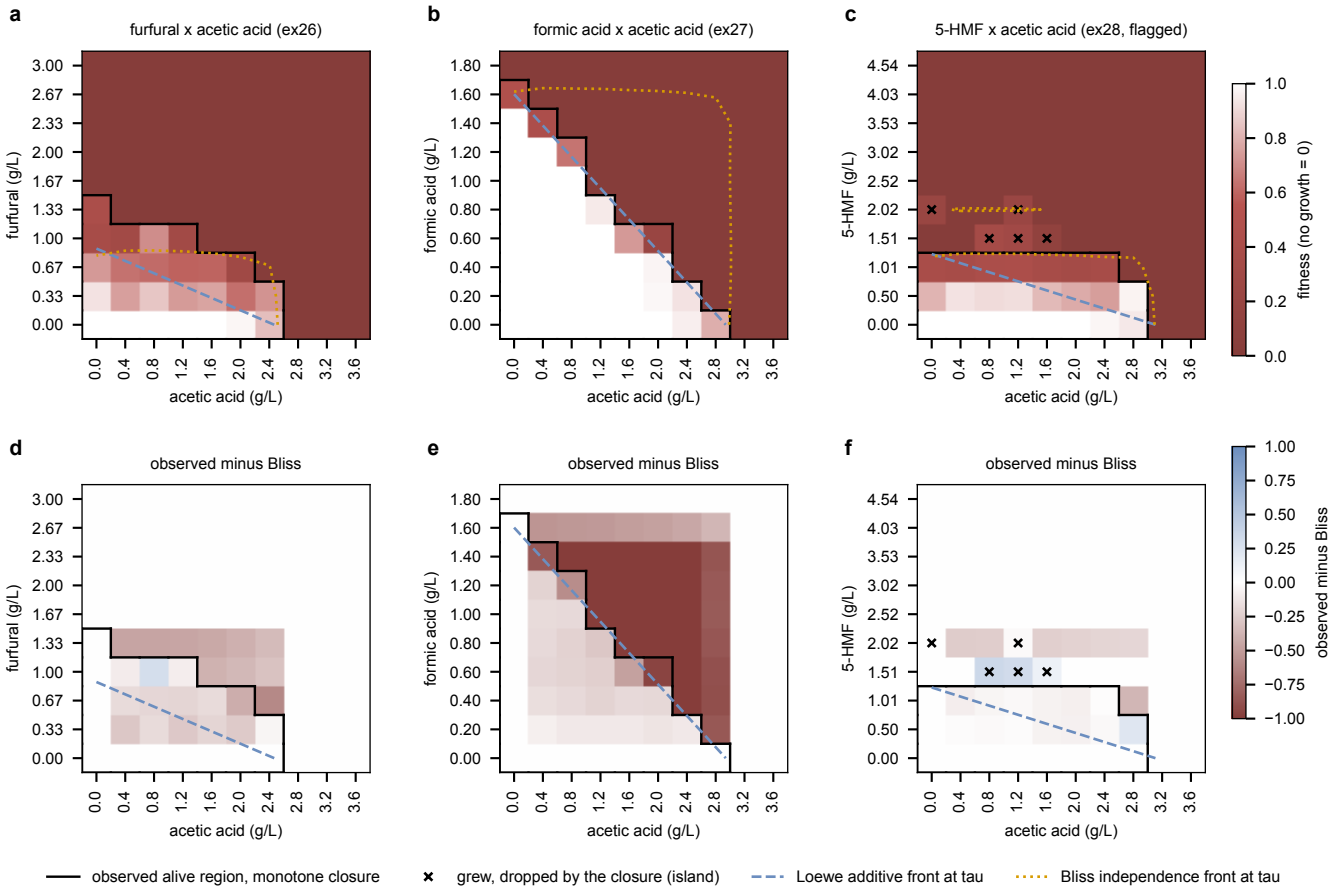


Figure 11: No single rule draws the boundary. **a** to **c**, Observed fitness over each isobole grid (no growth drawn as zero), the black outline around the monotone closure of the cells that grew, a cross on each grown cell the closure drops, and the two fronts at the grid's growth threshold: Loewe additivity dashed in blue, Bliss independence dotted in orange. Furfural with acetic acid follows Bliss, overlap 0.733 against 0.286 for Loewe, which calls 10 grown cells dead. Formic with acetic acid follows Loewe, overlap 0.875 against 0.436 for Bliss, which keeps 31 dead cells alive. 5-HMF with acetic acid follows Bliss, overlap 0.765 against 0.385. The growth edge along acetic acid tracks the winning rule within 0.444 (furfural), 0.333 (formic acid) and 0.778 (5-HMF) steps on average. ✗● **d** to **f**, Observed minus the empirical Bliss surface on a symmetric scale, with the Loewe front dashed. ✗●

Sources for fig. 11

a–c	`isobole_cells.csv` (observed, Bliss and Loewe per cell, the raw and the monotone growth call), `isobole_boundary.csv` (the overlap, the cells wrong each way and the edge error per grid, rule and observed region), `single_agent_fits.csv` (the axis Hill fits the Loewe front is drawn from). Served growth call; the isobole runs carry no software call. Status: on file.
d–f	`isobole_cells.csv`, `isobole_summary.csv` (the mean excess over Bliss and its bootstrap interval, printed in section 2). Status: on file.

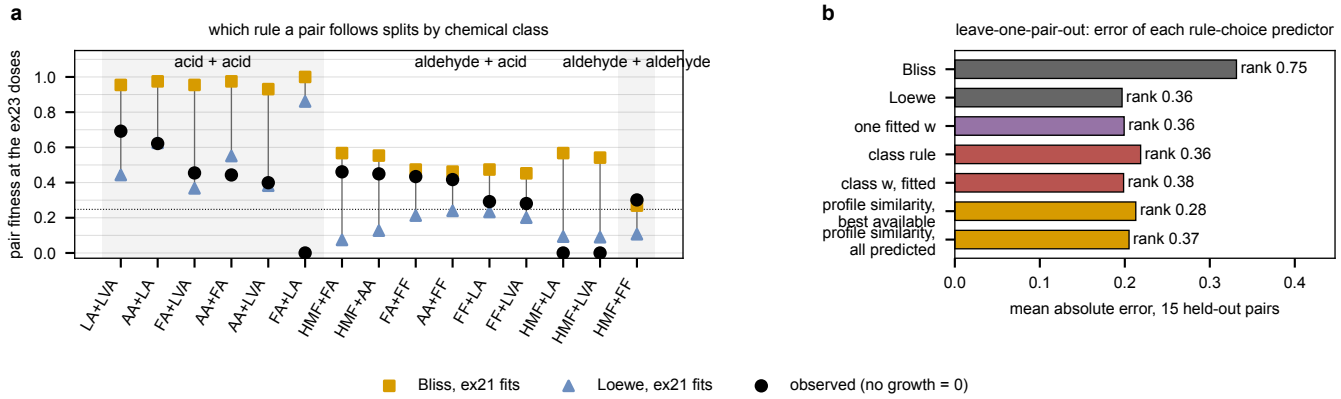


Figure 12: Which rule a pair follows, and whether anything predicts the choice. a, Each of the 15 fixed-dose pairs: observed fitness (no growth drawn as zero) against Bliss and Loewe from the titration fits, grouped by chemical class, the dotted line the growth threshold. All 6 acid pairs follow Loewe. The 8 aldehyde and acid pairs split 4 to 4, by the acid and not by the class: with acetic or formic acid the aldehyde follows Bliss, with lactic or levulinic acid it follows Loewe, and 5-HMF with either of those two grew in no well, which Loewe calls and Bliss does not. ✗● **b**, Leave-one-pair-out: each predictor gives a Loewe weight for the held-out pair, fitted on the other pairs, and the pair's fitness is that weight on Loewe plus its complement on Bliss. Mean absolute error over the held-out pairs, with the Spearman over the 12 grown pairs beside each bar. Pure Loewe has the lowest error, 0.197; pure Bliss the best ranking of the grown pairs, 0.748, because its error is a level bias that leaves their order intact. The growth call is not shown: with 12 of 15 pairs grown, Bliss scores 0.800 by predicting growth for every pair, a majority-class readout. A fitted global weight (0.199), the class rule (0.219), a fitted class weight (0.199) and profile similarity with the best available profiles (0.213) or the all-predicted ones (0.205) all land between the two rules. Nothing on file predicts the choice; with 15 pairs that is a small-sample statement. ✗●

Sources for fig. 12

a	`rule_choice_pairs.csv` from `mixture_combinations.csv` (served call) and `profile_similarity.csv`; the class of a pair is the chemical class of its two members. Status: on file.
b	`rule_choice_lopo.csv` and `rule_choice_lopo_pairs.csv`; the class hierarchy over all combinations (Loewe within a class, Bliss across) is in `rule_choice_higher_orders.csv`: growth AUROC 0.887 against Loewe's 0.958, with the smallest fitness error over the grown combinations (0.148 against 0.208). Status: on file.

What the figure now claims. The boundary is model-free and it is not one rule: Loewe for the acids, Bliss for the short acids with an aldehyde. The rule choice is the quantity a learned model would have to supply, and the leave-one-pair-out test is the bar it has to clear. That replaces "Loewe wins overall" as the statement of the bottom row.

The first model arms. The mixture trainer of this experiment ran its first round: the environment-encoder recipe of the corrected-store campaign with one compound token per inhibitor, a log-molar dose modulation (FiLM) that can be switched off, and a host head read at the mean over genes, fitted on the public dose anchors alone or on the anchors with the titration run, five compound-cold folds of one fold seed with three seeds each. The combination plate and the isobole grids are never trained on. 5 arms have score files here; two more, the dose-off pair, finished on Delta and await an rsync, and three are running there. Every arm sits below the model-free rules on every wet-lab readout.

Table 2: Round-one arms against the model-free bars, served growth call, median over the folds on file. The reference for growth is Loewe from the titration fits, AUROC 0.958; for the worst-scenario ranking over all combinations, Loewe, 0.772; for fitness over the combinations that grew, Bliss from the observed singles, 0.573; for the furfural grid, the mean of observed minus Bliss, -0.082; on the public side, nested ridge, 0.359. A negative grid residual means the arm predicts more growth than was seen.

arm	folds	growth AUROC	worst-scenario Spearman	fitness Spearman, grown	furfural grid, observed minus arm	public, centered Spearman
joint, anchors and titration, dose FiLM	5	0.475	-0.009	0.377	-0.751	0.261
anchors only, dose FiLM	5	0.295	-0.319	-0.346	-0.598	0.266
two-stage, head fine-tuned on the titration	5	0.556	0.093	0.216	-0.569	0.271
two-stage, head and environment encoder fine-tuned	5	0.582	0.154	0.188	-0.554	0.256
joint, two environment layers (still writing)	5	0.469	0.002	0.341	-0.765	0.252

What the arms say. The dose-aware joint arm fits the titration wells it trains on and then predicts the combinations at chance; the anchors-only arm, the zero-shot host, is below chance and ranks the grown combinations the wrong way. Switching the dose modulation off raises the growth AUROC in both host regimes (the Delta pair, recorded in the experiment note until its files arrive), which is the opposite of what a dose-aware model is for. Every arm's grid residual is negative: the arms predict more growth in the interior cells than was seen, and the dose-aware arms predict the most. Hypothesis (untested): the host head composes inhibitors near-additively or less, so each added inhibitor lowers predicted fitness by less than it lowers observed fitness, and the sharper the single-agent fit the larger that gap. On the public side every arm reads within a few hundredths of the earlier campaign's encoder and below ridge, so the wet-lab failure is not a model that was working elsewhere.

Where a learned model can still earn the panel. The rules need the single-agent curves. The regimes in which they have no prediction, or one they cannot improve with data, are the model's territory, and each needs its own baseline. Zero-shot singles: no Bioscreen well in training, the six titration curves predicted from public screens; the honest readout is the pooled ranking over all titration wells, since within one compound fitness falls with dose and any dose-aware model scores near perfectly there; on file the anchors arm reads weakly and the dose-off anchors arm reads nothing. One inhibitor unknown: five curves and the anchors known, the sixth withheld, the model predicts its curve and every combination that contains it; the baseline is the hybrid, the model's curve fitted as a Hill and fed to Loewe. Singles and pairs known, higher orders predicted: the fifteen pairs become training data and the forty-eight higher-order combinations the test; the baseline is a rule with one interaction parameter fitted on the pairs. Leave one pair out: the same, with the fifteenth pair as the test and the rule choice as the quantity learned.

Open decisions. First, whether the boundary readout, the overlap of alive regions and the edge error under the monotone closure, replaces growth AUROC as the primary wet-lab statistic of the bottom row; this document already scores both. Second, whether the fifteen pairs may become training data in the two regimes that need them; that breaks the pre-registered rule that the combination plate is never trained on, for those arms only, and the note would say so. Third, whether the bottom row ships with the model beside the rules, as the arm table shows it, or with the rules alone until an arm clears one of the bars. The first two are open; the third follows from whatever the queued rounds return.

Round three, read over both machines. The screen that followed round one (fig. 6) was read on 2026.10.12 with every arm scored on Delta and most of them on GilaHyper as well. Three readings. The same config at the same seed differs between the machines by a median of 0.065 and by up to 0.452 on the public score, so no single-seed difference in the screen is a finding. Over every run so far the one-, two- and three-source arms read 0.310, 0.311 and 0.305 on the public side and 0.567, 0.525 and 0.537 on the growth call: adding the public screens does not lift either readout, and the arms predict a cold compound on those screens at 0.062 (Hoepfner) and 0.029 (Hillenmeyer) against the chemistry-feature learners' 0.189 and 0.126. The dose modulation learns a held-out public dose (0.177) and lowers the growth call on 4 of 5 paired comparisons, which is round one's reading again under a protocol that selects on a held-out signal. Round four, six seeds on five folds of the three source-set arms, is what will turn the first reading into a measured null or a difference.

Rounds seven to ten, read over both machines. Round four returned the measured null: the three source-set arms at six seeds sit inside one spread on both readouts (fig. 6). Rounds seven to nine changed what a record is. Under the cell readout every record is a cell, the host the record with the empty genotype read through the shared head, and the public score did not move; what moved the private readout was the composition. With the compound tokens of a medium averaged, every arm read one fitness at one, two and three inhibitors (growth AUROC 0.48 to 0.57 over six runs); summed, the medium read fitter than its singles (0.27); read one inhibitor at a time and composed as Bliss independence, the same model orders the combinations at 0.862 fitted over 20 runs and 0.931 zero-shot, and its predicted fitness falls with the inhibitor count. Round ten (fig. 10) put three seeds on the questions. Sources: no public screen lifts the target, and the four-source arm is the lowest (0.265 to 0.194); the fitted wet-lab readout does not move with sources; the zero-shot readout does, 0.931, 0.951 and 0.944 on one, two and three sources, three runs each and the steps inside the spread. Dose: FiLM on the tokens reads the titration (0.928) and costs the host (0.644 against 0.862 with no channel), its public cost of the single-seed rounds being seed noise; the dose offset reads the titration at 0.869 at no public cost and composes at 0.851; the titration fine-tune with FiLM composes at 0.906.

What the public screens themselves can be predicted at is a narrower statement than it may read. The held-out compound score is the only thing measured on them: a compound the model never saw, its predicted profile across the HOP strains against the measured one, 0.058 over the round-ten arms against ridge's 0.189, and across the Hillenmeyer HET strains 0.010 against 0.126, the HET training loss rising rather than falling in every round. A seen compound at a held-out dose is a different question and is predicted (0.219). A held-out strain with every compound seen, the gene-cold split, has been run on Vanacloig alone, where it read near zero, and not on HOP or HET, so whether a new HOP or HET strain's response can be predicted is unmeasured rather than negative; the honest experiment is a gene-cold split on each screen scored as hit recovery over the strains called sick rather than a rank over every strain. Hoepfner's HIP arm is not in the pool at all, only its HOP arm and Hillenmeyer's HET arm. Hypothesis (untested): haploinsufficiency profiles are sparse, so a rank over all strains is a poor target for HET and a hit-level call would read differently.